

I225E Statistical Signal Processing

10. Least Squares Estimation

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Estimation Categories

Optimality Criterion:

- A) Classical estimation θ is nonrandom
 - Minimum variance unbiased estimation (MVUE)
 - Maximum likelihood estimation (MLE)
 - **Least squared estimation (LSE)**
- B) Bayesian estimation: θ is random
 - Minimum mean squared error (MMSE)

Least Squares Estimation

What if the data distribution is unknown?

⇒ **Least Squares Estimation**

[Features]

1. No probabilistic assumption on data
2. Assumption only on signal model
3. No guaranty for optimality
4. Performance cannot be assessed without assumption about probabilistic structure of data
5. Widely applied, due to its ease of implementation

LSE vs MLE

Metric	Least Squares Estimation (LSE)	Maximum Likelihood Estimation (MLE)
Requires Distribution?	No (Deterministic approach)	Yes (Probabilistic approach)
Primary Goal	Minimize the sum of squared residuals	Maximize the probability of seeing the data
Handling of Outliers	Sensitive (squares the errors)	Depends entirely on the chosen distribution

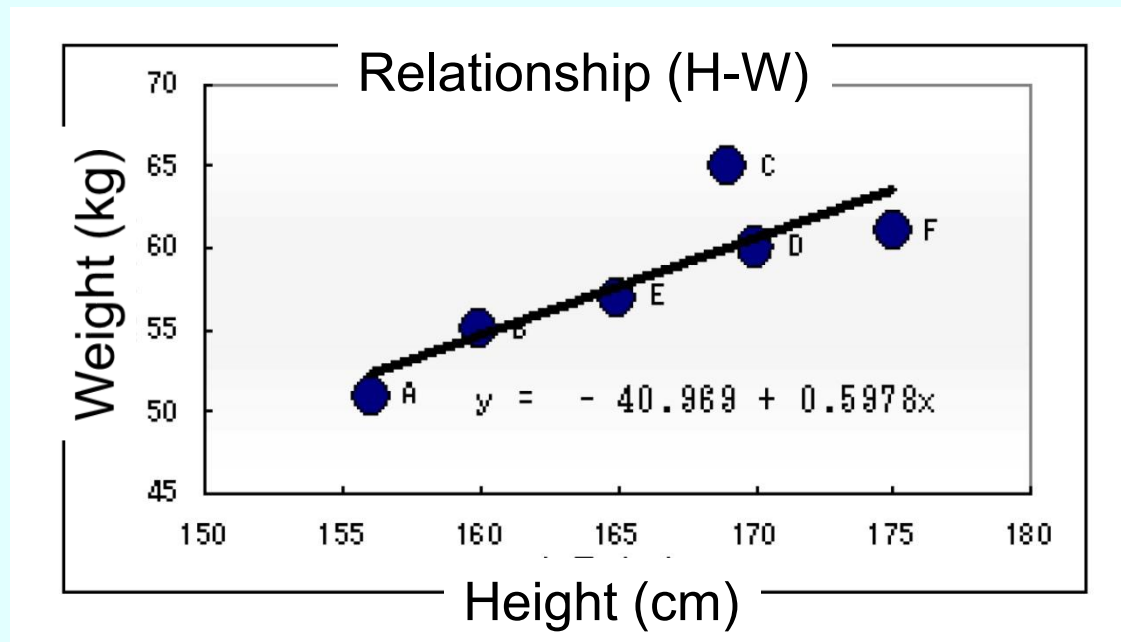
Least Squares Estimation

1. Linear least squares
2. Orthogonality principle
3. Matrix inversion lemma
4. Order-recursive least squares
5. Sequential least squares

Idea is simple:

With respect to observed data $s[n]$ and their estimation $\hat{s}[n; \theta]$, find coefficients θ of the estimation function in such a way that they minimize the squared error:

$$J(\theta) = \sum_{n=0}^{N-1} \{s[n] - \hat{s}[n; \theta]\}^2$$



1. Linear Least Squares

Goal: to estimate the unknown parameters of a linear model by **minimizing the sum of the squared differences (residuals)** between the observed data and the model's predictions.

If your system is $y = X\theta + e$, the solution to find the parameters θ is given by the following equation:

$$\theta = (X^T X)^{-1} X^T y$$

1. Linear Least Squares

Estimation function constructed by linear summation of observed data:

$$\hat{s}[n; \theta] = \sum_{m=1}^p H_{nm} \theta_m$$

In vector formula, $\hat{\mathbf{s}} = \mathbf{H}\theta$

- Linear coefficients: $\theta = [\theta_1, \theta_2, \dots, \theta_p]^T$
- Estimation: $\hat{\mathbf{s}} = [\hat{s}[0], \hat{s}[1], \dots, \hat{s}[N-1]]^T$
- Observation: $\mathbf{s} = [s[0], s[1], \dots, s[N-1]]^T$
- Observation matrix: $\mathbf{H} = |H_{nm}|_{n=0, \dots, N-1; m=1, \dots, p}$

Least squares error is

$$\begin{aligned} J(\theta) &= \sum_{n=0}^{N-1} (s[n] - \hat{s}[n; \theta])^2 \\ &= (\mathbf{s} - \mathbf{H}\theta)^T (\mathbf{s} - \mathbf{H}\theta) \\ &= \mathbf{s}^T \mathbf{s} - \mathbf{s}^T \mathbf{H}\theta - \theta^T \mathbf{H}^T \mathbf{s} + \theta^T \mathbf{H}^T \mathbf{H}\theta \\ &= \mathbf{s}^T \mathbf{s} - 2\theta^T \mathbf{H}^T \mathbf{s} + \theta^T \mathbf{H}^T \mathbf{H}\theta \end{aligned}$$

The least squares solution is

$$\begin{aligned} \frac{\partial J}{\partial \theta} &= -2\mathbf{H}^T \mathbf{s} + 2\mathbf{H}^T \mathbf{H}\theta = 0 \\ \Rightarrow \quad \hat{\theta} &= (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{s} \end{aligned}$$

If $\mathbf{H} \in \Re^{N \times p}$ is of full rank p , $\mathbf{H}^T \mathbf{H}$ is invertible and the least squares solution can be obtained.



The least squares error is

$$\begin{aligned} J_{min} &= J(\hat{\theta}) \\ &= (\mathbf{s} - \mathbf{H}\hat{\theta})^T (\mathbf{s} - \mathbf{H}\hat{\theta}) \\ &= (\mathbf{s} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{s})^T (\mathbf{s} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{s}) \\ &= \mathbf{s}^T (\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T)^T (\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T) \mathbf{s} \\ &= \mathbf{s}^T (\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T) \mathbf{s} \end{aligned}$$

This can be further simplified as

$$\begin{aligned} J_{min} &= \mathbf{s}^T \mathbf{s} - \mathbf{s}^T \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{s} \\ &= \mathbf{s}^T (\mathbf{s} - \mathbf{H}\hat{\theta}). \end{aligned}$$

Exercise

A researcher is monitoring a sensor that measures a signal $s[n]$ changing over discrete time steps n . Based on the physical properties of the system, the researcher models the true signal as a linear function of time:

$$\hat{s}[n; \theta] = \theta_1 + \theta_2 n$$

The researcher collects three data points ($N = 3$) at times $n = 0, 1, 2$: $s[n] = [1, 2, 4]$

- a) Formulate the matrix equation
- b) Compute the Least Squares Estimate
- c) Compute the final scalar value of the minimum cost function $J(\hat{\theta})$

Answer

- a) Formulate the matrix equation

$$s = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, H = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

- a) Compute the Least Squares Estimate

$$\hat{s}[n] = \frac{5}{6} + 1.5n$$

- a) Compute the final scalar value of the minimum cost function $J(\hat{\theta})$

$$J(\hat{\theta}) = \frac{1}{6}$$

2. Orthogonality Principle

- Orthogonality principle is a geometric concept that provides a powerful shortcut for **finding the optimal least squares solution**.
- It states that **the estimation error vector must be orthogonal (at a 90° angle)** to the subspace spanned by your input data.

Decomposing the observation matrix $\mathbf{H}$ into column vectors $\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_p$.

$$\mathbf{s} = [\mathbf{h}_1 \ \mathbf{h}_2 \ \dots \ \mathbf{h}_p] \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix} = \sum_{i=1}^p \theta_i \mathbf{h}_i$$

Error vector is

$$\boldsymbol{\epsilon} = \mathbf{s} - \mathbf{H}\boldsymbol{\theta} = \mathbf{s} - \sum_{i=1}^p \theta_i \mathbf{h}_i$$



The least squares error and the local minimum condition are

$$\begin{aligned} J(\theta) &= \|\epsilon\|^2 = \|\mathbf{s} - H\theta\|^2 = \left\| \mathbf{s} - \sum_{i=1}^p \theta_i \mathbf{h}_i \right\|^2 \\ \frac{\partial J}{\partial \theta_i} &= -2\mathbf{h}_i^T (\mathbf{s} - \sum_{i=1}^p \theta_i \mathbf{h}_i) \\ &= -2\mathbf{h}_i^T \epsilon = 0 \\ &\rightarrow \epsilon \perp \mathbf{h}_i \end{aligned}$$

If coefficients θ_i are chosen in such a way that error vector ϵ is orthogonal to observation vectors $\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_p$, the least squares error J is minimum.

Exercise

An engineer is trying to estimate a 3D signal vector, $\mathbf{s} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$.

However, their measurement device can only observe data along a 2D plane spanned by two basis vectors, $\mathbf{h}_1$ and $\mathbf{h}_2$:

$$\mathbf{h}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{h}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

The estimated signal must be a linear combination of these two vectors: $\hat{\mathbf{s}} = \theta_1 \mathbf{h}_1 + \theta_2 \mathbf{h}_2$.

- a) Apply the orthogonality principle
- b) Calculate the optimal parameters $\hat{\theta}_1$ and $\hat{\theta}_2$.
- c) Calculate the final error vector ϵ and compute its dot product with both $\mathbf{h}_1$ and $\mathbf{h}_2$ to prove they are perpendicular.

Inversion of partitioned matrix

- $(p+q) \times (p+q)$ matrix $\mathbf{M}$ partitioned as:

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \mathbf{A} \in \mathbb{R}^{p \times p}, \mathbf{B} \in \mathbb{R}^{p \times q}, \mathbf{C} \in \mathbb{R}^{q \times p}, \mathbf{D} \in \mathbb{R}^{q \times q}$$

- Inversion of partitioned matrix:

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^{-1} = \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} & -\mathbf{A}^{-1}\mathbf{B}(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} & (\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \end{pmatrix}^{-1}$$

- (Exercise) Prove by:

$$\mathbf{M}^{-1}\mathbf{M} = \mathbf{M}\mathbf{M}^{-1} = \mathbf{I}_{p+q}$$

3. Matrix inversion lemma

■ Suppose matrices:

$$\mathbf{A} \in \mathbb{R}^{p \times p}, \mathbf{B} \in \mathbb{R}^{p \times q}, \mathbf{C} \in \mathbb{R}^{q \times q}, \mathbf{D} \in \mathbb{R}^{q \times p}$$

■ Matrix inversion lemma

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1}$$

■ (Exercise): Prove the lemma by showing:

$$(\mathbf{A} + \mathbf{BCD})(\mathbf{A} + \mathbf{BCD})^{-1} = (\mathbf{A} + \mathbf{BCD})^{-1}(\mathbf{A} + \mathbf{BCD}) = \mathbf{I}_p$$

4. Order-Recursive Least Squares

Denote observation matrix as $\mathbf{H}(k) (\in R^{N \times k})$ and observed data as $s (\in R^N)$. k th order least square estimation based on $\mathbf{H}(k)$ is given by

$$\hat{s}[n] = \sum_{m=1}^k H_{nm}(k) \theta_m(k).$$

Suppose a column is added to the observation matrix

$$\mathbf{H}(k+1) = [\mathbf{H}(k) \quad \mathbf{h}_{k+1}] = [N \times k \quad N \times 1]$$

Then, $(k+1)$ th order least square estimation

$$\hat{s}[n] = \sum_{m=1}^{k+1} H_{nm}(k+1) \theta_m(k+1)$$

can be updated via the following recursive formula

$$\hat{\theta}(k+1) = \left[\begin{array}{c} \hat{\theta}(k) - \frac{\{H^T(k)H(k)\}^{-1}H^T(k)\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{s}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ \frac{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{s}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \end{array} \right] =$$

$\begin{bmatrix} k \times 1 \\ 1 \times 1 \end{bmatrix}$

Here, $\mathbf{P}_k^\perp = \mathbf{I} - \mathbf{H}(k)\{\mathbf{H}^T(k)\mathbf{H}(k)\}^{-1}\mathbf{H}^T(k)$.

The minimum least square error is updated as

$$J_{min}(k+1) = J_{min}(k) - \frac{(\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{s})^2}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}}$$

Moreover, inverse matrix, denoted as $\mathbf{D}_k = (\mathbf{H}^T(k)\mathbf{H}(k))^{-1}$, $\mathbf{D}_k$ can be updated as follows

$$\mathbf{D}_{k+1} = \begin{bmatrix} \mathbf{D}_k + \frac{\mathbf{D}_k \mathbf{H}^T(k) \mathbf{h}_{k+1} \mathbf{h}_{k+1}^T \mathbf{H}(k) \mathbf{D}_k}{\mathbf{h}_{k+1}^T \mathbf{P}_k^\perp \mathbf{h}_{k+1}} & - \frac{\mathbf{D}_k \mathbf{H}^T(k) \mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T \mathbf{P}_k^\perp \mathbf{h}_{k+1}} \\ - \frac{\mathbf{h}_{k+1}^T \mathbf{H}_k \mathbf{D}_k}{\mathbf{h}_{k+1}^T \mathbf{P}_k^\perp \mathbf{h}_{k+1}} & \frac{1}{\mathbf{h}_{k+1}^T \mathbf{P}_k^\perp \mathbf{h}_{k+1}} \end{bmatrix}$$

[Procedure]

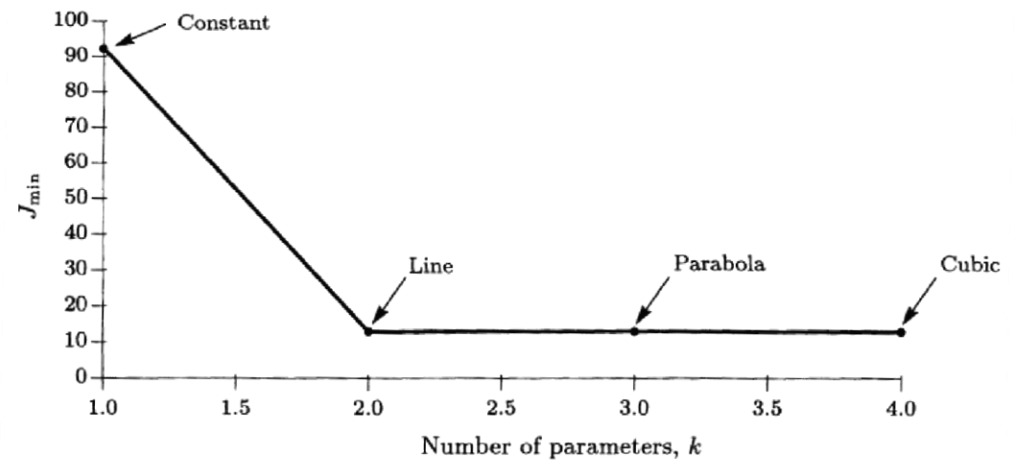
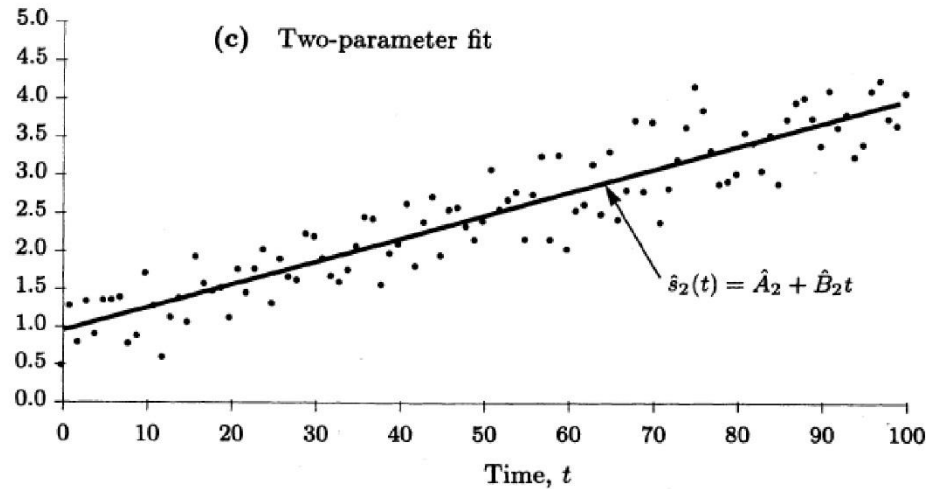
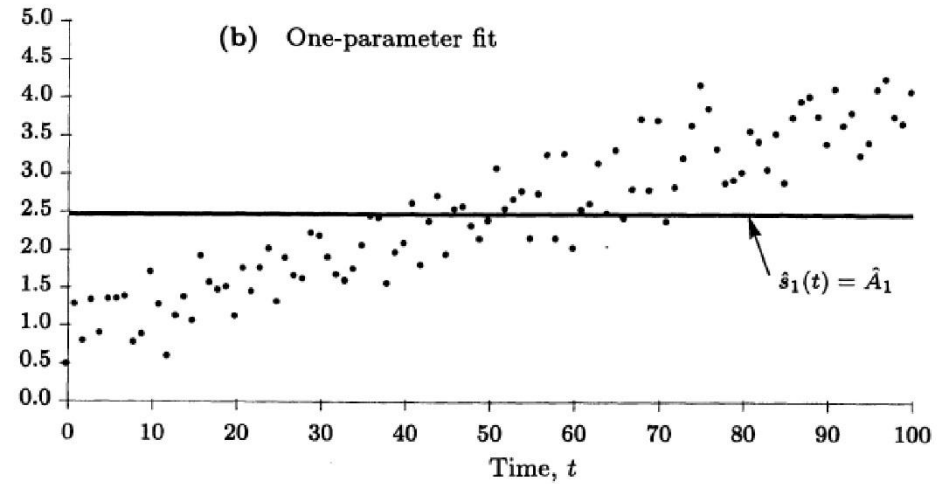
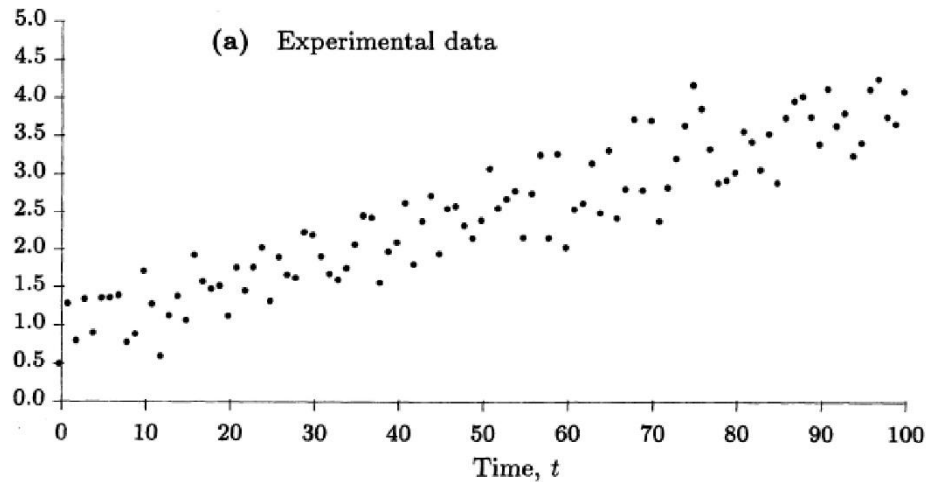
1. If $k = 1$,

$$\hat{\theta}(1) = \{\mathbf{H}^T(1)\mathbf{H}(1)\}^{-1}\mathbf{H}^T(1)\mathbf{s}$$

$$J_{min}(1) = \{\mathbf{s} - \mathbf{H}(1)\hat{\theta}(1)\}^T \{\mathbf{s} - \mathbf{H}(1)\hat{\theta}(1)\}$$

2. As the order is increased ($k \rightarrow k + 1$), $\hat{\theta}(k)$, $J_{min}(k)$, D_k are updated accordingly.

Example: Curve Fitting



5. Sequential Least Squares

For observation data $s(n-1) (\in R^n)$ and observation matrix $H(n-1) (\in R^{n \times p})$, consider the linear estimation:

$$\hat{s} = H(n-1)\theta.$$

We denote the least square solution as $\hat{\theta}(n-1)$. Suppose that additional data is observed as

$$s(n) = [s[0] \ s[1] \ \cdots \ s[n-1] \ s[n]]^T$$

$$H(n) = \begin{bmatrix} H(n-1) \\ h^T(n) \end{bmatrix} = \begin{bmatrix} n \times p \\ 1 \times p \end{bmatrix}$$



Then, the least square solution $\hat{\theta}(n)$ can be updated sequentially as follows.

Estimator Update:

$$\hat{\theta}(n) = \hat{\theta}(n-1) + \mathbf{K}(n)\{s[n] - \mathbf{h}^T(n)\hat{\theta}(n-1)\}$$

$$\mathbf{K}(n) = \frac{\boldsymbol{\Sigma}(n-1)\mathbf{h}(n)}{\sigma_n^2 + \mathbf{h}^T(n)\boldsymbol{\Sigma}(n-1)\mathbf{h}(n)}$$

Variance Update:

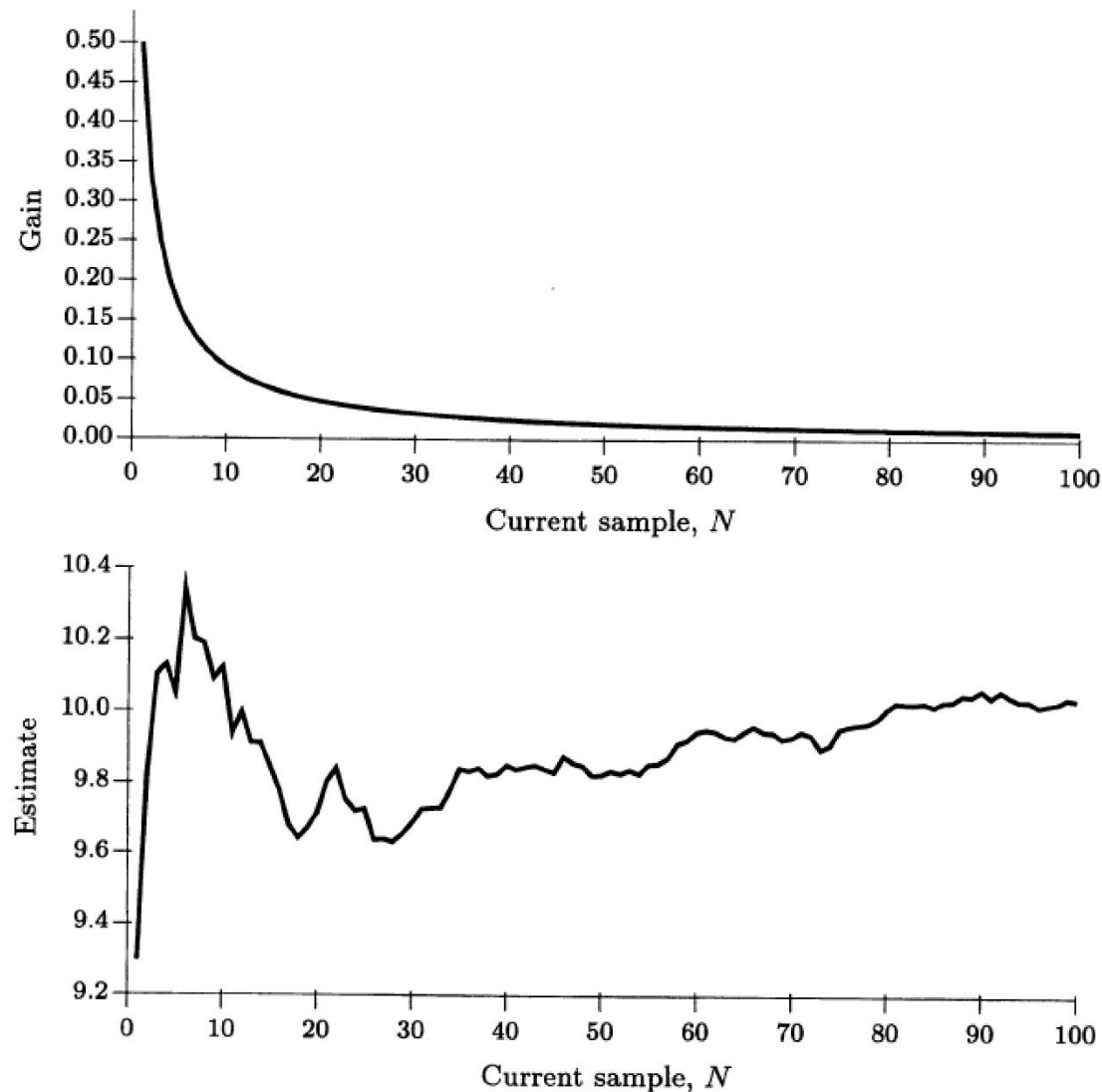
$$\boldsymbol{\Sigma}(n) = \{\mathbf{I} - \mathbf{K}(n)\mathbf{h}^T(n)\}\boldsymbol{\Sigma}(n-1)$$

Error Update:

$$J_{min}(n) = J_{min}(n-1) + \frac{\{s[n] - \mathbf{h}^T(n)\hat{\theta}(n-1)\}^2}{\sigma_n^2 + \mathbf{h}^T(n)\boldsymbol{\Sigma}(n-1)\mathbf{h}(n)}$$

- Noise $s - \hat{s}$ are independent; their covariance matrix has $\text{diag}(\sigma_0^2, \sigma_1^2, \dots, \sigma_n^2)$.
- Σ stands for covariance matrix of $\hat{\theta}$ ($\Sigma = \mathcal{C}_\theta$).
- No need to compute inverse matrix.
- On-line process is possible.

Sequential least squares for DC level in white Gaussian noise



Exercise

Imagine you are tracking a constant scalar parameter θ over time using a sensor.

- At Time Step $n - 1$, your parameter estimate is $\hat{\theta}(n - 1) = 10$.
- Your uncertainty matrix (which is just a scalar variance here) is $\Sigma(n - 1) = 2$.
- The sensor noise variance is known to be $\sigma_n^2 = 1$.

Suddenly, at Time Step n , a new observation vector arrives: $\mathbf{h}(n) = 1$. The sensor outputs a value of $s[n] = 13$.

- a) Calculate the Gain Vector $\mathbf{K}(n)$
- b) Update the Parameter Estimate $\hat{\theta}(n)$
- c) Update the Uncertainty Matrix $\Sigma(n)$